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MILITARY SPECIFICATION

TABLES, ROTARY, HORIZONTAL, MANUAL AND POWER FEED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers rotary tables for mounting and rotating work and locating it in precise angular positions. Indexing manual feed and power feed rotary tables are included with T-slots for workholding. Tables included are intended for use as either primary or auxiliary work tables.

1.2 Classification. The rotary tables shall be of the following types, styles, and sizes. The type, style, and size to be furnished shall be as specified (see 6.2.1).

Type I - Manual hand revolving rotary tables with lift assistance.

Type II - Precision worm and worm gear drive positioning with manual handwheel.

Type III - Motorized positioning system with either an AC or DC motor drive system.

Type IV - Feeding table gear change system with either an AC or DC motor drive system.

Type V - Feeding table with variable speed DC motor drive system.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, Memphis, TN 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Style A - Round configuration.
Size 1 - 24 inches (610 mm) diameter.
Size 2 - 30 inches (762 mm) diameter.
Size 3 - 36 inches (914 mm) diameter.
Size 4a - 42 inches (1067 mm) diameter.
Size 5 - 48 inches (1219 mm) diameter.
Size 6 - 60 inches (1524 mm) diameter.
Size 7 - 72 inches (1829 mm) diameter.

Style B - Square or Rectangular Configuration.
Size 1 - 24 x 24 inches (610 x 610 mm) dimension.
Size 2 - 30 x 30 inches (762 x 762 mm) dimension.
Size 3 - 36 x 36 inches (914 x 914 mm) dimension.
Size 4a - 36 x 50 inches (914 x 1270 mm) dimension.
Size 5a - 42 x 42 inches (1067 x 1067 mm) dimension.
Size 6 - 48 x 48 inches (1219 x 1219 mm) dimension.
Size 7 - 48 x 60 inches (1219 x 1524 mm) dimension.
Size 8a - 48 x 72 inches (1219 x 1829 mm) dimension.
Size 9 - 60 x 60 inches (1524 x 1524 mm) dimension.
Size 10 - 60 x 72 inches (1524 x 1829 mm) dimension.
Size 11 - 60 x 84 inches (1524 x 2134 mm) dimension.
Size 12 - 72 x 72 inches (1829 x 1829 mm) dimension.
Size 13 - 72 x 84 inches (1829 x 2134 mm) dimension.
Size 14 - 72 x 96 inches (1829 x 2438 mm) dimension.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Service.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

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MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

2.1.2 Other Government documents. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

29 CFR, CHAPTER XVII, PART 1910 - Occupational Safety and Health Standards.

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B5.1 - T-Slots, Their Bolts, Nuts, and Tongues.

ANSI B5.10 - Machine Tapers.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018-3363.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103-1187.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA ICS 1 - Industrial Controls and Systems.

NEMA MG 1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037-1570.)

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NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The rotary tables shall be new and one of the manufacturer's current models capable of mounting either on primary machine tables or on any rigid base suitably prepared for such use in accordance with the requirements specified herein. The rotary table shall include all components, parts, and features necessary to meet the performance requirements specified herein. Additional features which are not specified but are a part of the manufacture's current design shall be included in the rotary tables furnished. Rotary tables shall be designed for rotary motion 360 degrees clockwise and counterclockwise. The rotary tables may be manually positioned by (a) hand operated gear, (b) manually positioned with air lift assistance, (c) positioned by motor and worm gear, (d) power fed at variable speeds with pick-off gears, (e) power fed by variable speed, silicon control rectifier (SCR) direct current DC electronic drive, or (f) positioned manually or by power with hydrostatic assistance. Rotary tables shall be a self-contained unit, capable of being used independently as an inspection instrument, or as a workholding device for use in varied machining operations. Electrical and hydraulic power systems, when specified for performance of specified functions, shall be complete and integral parts of the equipment. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair. Rotary tables and all its components parts shall be designed to operate satisfactorily in ambient temperatures up to 112 degrees Fahrenheit.

3.2.1 Dimensional system. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the rotary table. When only one dimensional system is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all dimensions, measurements,

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sizes, and capacities are given in US units. These dimensions may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, depth stops, carriage stops, dial indicators, pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the rotary tables shall be graduated in the same system.

3.2.2 Reclaimed materials. The rotary tables may contain reclaimed materials provided such materials will not jeopardize the rotary tables intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the rotary tables.

3.2.3 Energy efficiency. The rotary tables and its components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Controls. All operating controls shall be located conveniently to the operator at his normal work station.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the rotary table that present safety hazards. The safety devices shall not interfere with the operation of the rotary table. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All parts, components, mechanisms, and assemblies furnished on the rotary table, whether or not specifically required herein, shall comply with all of the requirements of 29 CFR, Chapter XVII, Part 1910 that are applicable to the rotary table itself. The rotary table shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer's commercial rotary table supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required because the guard configuration is dependent on the size and configuration of the workpices. The above, and any other additional safety and health requirement should be specified in detail (see 6.2.1).

3.2.6 Mercury restriction. The rotary tables shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the rotary tables.

3.2.8 Environmental protection. The rotary table shall be so designed and constructed that, under the operating, service, transportation, and stor-

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age conditions described herein, the rotary tables shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation.

3.2.9 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a cleanable or replaceable filter. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.10 Interchangeability. All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or rotary tables.

3.3 Construction. The rotary tables shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the rotary tables at its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the rotary tables. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the rotary tables shall be painted in accordance with the manufacturer's standard commercial practice and color.

3.3.6 Threads. All threaded parts used on the rotary tables and its related attachments and accessories shall conform to FED-STD-H28.

3.3.7 Dials. All dials that indicate the rotary tables' motion shall be graduated in .01 degrees per indexing step of motion. Dials that indicate pounds per square inch (psi), or input air pressure shall be such that graduations may be read from the normal operating position. Dial type mode switches shall be marked with captions which show clearly and legibly the type

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function to be performed. Dials shall be permanently and legibly engraved on a non-glare background.

3.3.8 Gears. All gears shall be constructed from a material suitable for the intended purpose. All drive gears shall have the hardness, surface finish, and toughness that will enable the gear train to transmit full rated torque of the drive motor without gear damage, failure, or premature deterioration and wear.

3.3.9 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirement shall be as specified.

3.4 Components. The rotary table shall consist of, but not be limited, to a worktable, base bearing support, scales, coolant system, and, when required, a hydraulic system and a pneumatic system.

3.4.1. Worktable. The worktable shall be constructed of close grained cast iron or alloy steel of high tensile strength and shall be braced and reinforced. It shall be capable of supporting and rotating the maximum table capacities for on center and off-center loads when cutting forces are applied. The number and size of the T-slots shall be in accordance with tables I or II and all T-slots shall conform to the requirements of ANSI B5.1. T-slots of each table shall be parallel with each other within the alignment tolerance of table III. Unless otherwise specified (see 6.2.1), the round table top shall be provided with either parallel or radial T-slots. The center of the table, when required, shall be provided with a precision taper in accordance with the requirements of ANSI B5.10, or a centering hole of the size and type specified (see 6.2.1). The center taper shaft shall be rigidly supported in a precision type ball or roller bearing for radial loads. As required, circular ways shall be provided to furnish bearing support for the periphery of the table top during rotation or in the indexing position. The ways of tables with feeding capability shall be designed in combination with bearing thrust support to prevent lifting of the table by working forces. A manual clamping system shall be provided for tables with load capacities of 27,000 pounds or less and a pneumatic or power clamping system shall be provided for tables with load capacities of 27,001 pounds or more. The table top shall be provided with drilled and tapped holes to permit mounting of jig fixtures with a variety of bolt patterns. The table shall operate over a range of loads from 1 lb to 100 lbs with no loss of indexing accuracy.

3.4.2 Base. The bottom of the base shall be precision finished for mounting on and securing to a flat and level surface. The construction shall be sufficiently rigid to sustain the weight and working forces for which the table is designed. The rotating base shall function as a support for all components of the table and shall provide 360 degrees rotation of the table. A large diameter anti-friction bearing shall be provided for styles A and B style tables which shall permit rotation of the table with minimum friction and shall serve as a thrust bearing to prevent lifting of the table under working forces. If ways integral with the table top are matched with the base, the matching ways shall be machined and scraped or precision ground. When specified (see 6.2.1), cast iron ways, if used, shall be flame hardened to a minimum hardness equivalent to 45 on the Rockwell "C" scale.

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3.4.3 Bearing supports. The center of the table shall be radially supported on precision type bearings. Feeding tables of all types shall be fitted with large diameter pre-loaded anti-friction bearings which will prevent axial and radial movement under working loads.

3.4.4 Encoder. An encoder or digital readout system , with graduations of not greater than one degree intervals, and an inductosyn scale usable for positioning to within plus or minus 10 seconds of arc shall be provided to indicate the angular position of each manually operated table. All tables, including those with digital readout system, shall be provided with suitable means for determining table position.

3.4.5 Coolant system. The table shall be supplied with a complete flood and mist type coolant system with an automatic controlled selection of the amount, coolant type, and on/off of coolant flow. The system shall have adequate capacity for all types of rotary table operations for which the rotary table is designed and shall have means for regulating the flow of flood coolant and for adjusting mist coolant supply as required. All components necessary to make the system completely functional shall be furnished including reservoirs, chip screens, filters, drain troughs, and clean-out plates.

3.4.6 Hydraulic system. When furnished, the hydraulic system, shall be complete with pump, valves, piping, cylinders, reservoir, and pressure controls. Overpressure protection shall be provided on the high pressure line from the hydraulic pump to prevent damage to components. The system shall include filters to ensure delivery of clean fluid. The reservoir shall have a sight gage to indicate the fluid level. The fluid temperature shall be maintained within the proper operating range during all normal operations of the table.

3.4.6.1 Pneumatic system. When required (see 3.4.1), a pneumatic system shall be provided and shall operate on shop air having a line pressure of 70 to 90 psi. The pneumatic system shall have an air line filter, a water trap, and an oil mist lubricator, as necessary. The system shall include all devices such as pressure regulators, pressure reducers, solenoid valves, and check valves that are necessary to meet the requirements specified herein. Air pressure gauges shall be provided to indicate input air pressure in all lines that are downstream from a pressure control device.

3.4.7 Table rotation. Table rotation shall be by one of the methods described in 3.4.7.1 through 3.4.7.5. Provision shall be made for clamping the table at intermediate positions and for holding the table firmly in clamped positions without distortion of the mounting surface by workload or by cutting forces. All rotating tables shall be capable of being indexed in 1 degree increments over the entire 360 degree range. Indexing accuracy when rotated from one locked position to any other locked position shall be within 0.25 arc seconds.

3.4.7.1 Type I manual hand revolving rotary table with lift assisting. When specified (see 6.2.1), the type I rotary table shall be provided with either, an air-oil mist assistance, hydraulic assistance, air bearing assistance, or fluid pressure assistance to minimize friction between the base and table ways during table index movement. The type I rotary table shall be

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provided with a minimum center load capacity of not less than that specified in tables I and II. Regardless of the system used between the base and table, the lift assistance shall not be lost due to unbalance workload on the table. Each manual rotary table shall be provided with a handwheel and inductosyn scale or optical encoder with digital readout for positioning to plus or minus 10 seconds of arc.

3.4.7.2 Type II precision worm and worm gear and optical encoder positioning system. When specified see 6.2.1, the type II rotary table shall be provided with a precision worm and worm gear drive positioning system and optical encoder with digital readout. The precision worm and worm gear drive positioning system shall be capable of moving the table manually in both directions to within plus or minus 10 seconds of arc to the desired positioning. A means of adjustment for backlash shall be provided which will permit adjustment for gear wear without major disassembly. The type II rotary table shall be provided with a minimum center load capacity of not less than that specified in tables I and II.

3.4.7.3 Type III AC or DC motorized positioning system. When specified (see 6.2.1), the type III rotary table shall be provided with a reversible motor positioning drive system. The type III rotary table shall be provided with a reversible variable speed direct current (DC) motor drive or single speed alternate current (AC) gear motor drive for positioning the table with reduction gearing that shall be provided with a positioning speed of not less than 100 inches per minute (ipm) on the periphery of the table. The type III rotary table shall be provided with a minimum center load capacity of not less than that specified in tables I and II. A manual handwheel or crank shall be provided for precise final positioning of the table.

3.4.7.4 Type IV with AC or DC motor drive system feeding tables, gear change system. When specified (see 6.2.1), the type IV rotary table shall be provided with a gear box which shall provide not less than six feeding speeds. The tables feed range shall be as specified in tables I and II. A rapid traverse mechanism independent of the pick-off gears shall be provided for table positioning. When specified (see 6.2.1), a manual shift type gear change shall be supplied in lieu of pick-off gear. Each type IV rotary table shall be provided with a minimum center load capacity of not less than that specified in tables I and II.

3.4.7.5 Type V feeding tables, variable speed DC motor drive. When specified (see 6.2.1), the type V rotary table shall be provided with a reversible, infinitely variable speed, direct current (DC) motor drive. The rotary motion shall be provided with either adjusting sine or anti-backlash dual pinion and sine ring gear powered by a DC motor with reversing controls and shall be provided with a feed range variable from not less than that specified in tables I and II. Each rotary table shall be provided with a minimum center load capacity of not less than that specified in tables I and II.

3.4.8 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to NFPA 79. Each rotary table shall draw all of its electrical power from a single 230/460-volt, 3-phase, 60-Hz circuit. Each rotary table shall have a fused safety disconnect switch or circuit breaker. The rotary table shall be initially wired for operation on 460 volts. An identified terminal for grounding the rotary table when it is installed

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shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in NFPA 79 for the disconnect fuse rating. Enclosed type speed controls shall be mounted in an enclosure conforming to NEMA ICS 1.

3.4.8.1 Digital readout system. When specified (see 6.2.1), a digital readout system shall be installed on the rotary table axis of travel. The digital readout system shall consist of a position feedback device, an electronic display for each axis specified, and all related components, mounting hardware, and electrical equipment necessary to assure complete operation of the system. The system shall be capable of providing a readily visible digital display over the full working range of each axis on which installed. The system resolution shall be 0.001 degree when specified. The system accuracy shall be within plus or minus 0.002 degrees and the repeatability shall be plus or minus 10 seconds of a table arc. Special requirements for the digital readout system components shall be as specified (see 6.2.1). The readout system shall be capable of indicating axis locations in both US and SI systems of measurements.

3.4.8.2 Position feedback device. A position feedback device including all related components, hardware, and electrical equipment shall be provided for each axis having a readout system. The position feedback device shall be the independent type. The circular interpolation position device for each axis specified shall be measured by a 720-Pole 12 inch inductosyn scale or resolver feedback system. The feedback shall be provided with a positioning accuracy to within plus or minus 10 seconds of table arc. The device shall provide axis circular interpolation feedback rotary motion bi-directional of any arc between 0 and 360 degrees over the entire range of the axis rotary motion for each axis specified. The feedback device shall be installed with a cover or guard to prevent physical damage to any portion or the feedback device during operation of the rotary table.

3.4.8.3 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for a dripproof enclosure conforming to NEMA-MG1.

3.4.8.4 Controls. Unless otherwise specified (see 6.2.1), the rotary table shall be equipped with a remote control station. Means shall be provided for the operator to unclamp the top of the rotary table and energize the air-oil mist system. A positioning system shall be provided for the operator to rotate the table to the desired index position with a sensing dog or other manufacturer system. The system shall direct the rotary table to the desired position and shall actuate the positioning system to provide the following:

a. Provisions to move the shot pin to index stop.
b. Button or switch to de-energize the air-mist system.
c. Button or switch to clamp and unclamp table top.
d. Provisions for retracting the table from index stop.

In addition to speed and convenience, the automatic indexing cycle shall provide a consistent sequence to assure that the locating pin is retracted during

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TABLE I. Style A, round table, size and capacity.

Rotary table size	Rated work-load 1/ pounds	Table diameter inches	Table height inches	Number of parallel T-slots where applicable	Number of radial T-slots where applicable	T-Bolt size inch	Horsepower of positioning motors	Horsepower of feeding motors	Table rotation feeding range variable rpm
1	10000	24	10	6	4	1/2	1/2	3/4	0 to 2
2	10000	30	11	6	5	1/2	9/16	1	0 to 2
3	15000	36	11	6	5	5/8	9/16	1	0 to 2
4a	45000	42	11	6	5	5/8	5/8	2	0 to 2
5	50000	48	13	8	5	5/8	5/8	3	0 to 1
6	50000	60	14-18	8	7	3/8	5/8	5	0 to 1
7	75000	72	14-18	8	7	5/8	3/4	5	0 to 1

Requirements are not less than those shown. Where a range is shown, it shall encompass a range from the stated minimum or less to the stated maximum or greater.
1/ with workpiece centered on table.

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TABLE II. Style B, square or rectangular table, size and capacity.

Rotary table size	Rated work-load 1/ pounds	Table diameter inches	Table height inches	Number of parallel T-slots where applicable	T-slots size inch	Horsepower of positioning motors	Horsepower of feeding motors	Table rotation feeding range variable rpm
1	10000	24 x 24	10	6	1/2	1/3	3/4	0 to 2
2	10000	30 x 30	10	6	1/2	1/3	3/4	0 to 2
3	15000	36 x 36	11	6	5/8	1/2	3/4	0 to 2
4a	20000	36 x 50	11	6	5/8	1/2	3/4	0 to 2
5a	25000	42 x 42	11	6	5/8	1/2	1	0 to 2
6	28000	48 x 48	12-13	6	5/8	3/4	2	0 to 1
7	50000	48 x 60	12-13	6	5/8	1	2	0 to 1
8	50000	48 x 72	12-13	8	5/8	1	2	0 to 1
9	50000	60 x 60	12-18	8	5/8	2	3	0 to 1
10	50000	60 x 72	12-18	8	5/8	2	3	0 to 1
11	75000	60 x 84	16-18	8	1 1/16	2	5	0 to 1
12	75000	72 x 72	16-18	8	1 1/16	2	7	0 to 1
13	75000	72 x 84	16-18	8	1 1/16	2	7	0 to 1
14	75000	72 x 96	16-18	8	1 1/16	2	7	0 to 1

Requirements are not less than those shown. Where a range is shown, it shall encompass a range from the stated minimum or less to the stated maximum or greater.
1/ with workpice centered on table.

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TABLE III. Alignment tolerances.

Measurement or determination	Maximum allowable deviation, not more than, inches
T-slots parallel with reference slot	0.0003 per one foot span 0.001 total
Table top concentricity from one lock position to any other shall be within	0.000050 inches TIR
Indexing accuracy from one locked to another shall be within	0.24 arc seconds total
Parallelism work surface of table top to the base	0.0001 inch TIR
Flatness and finish working surface of the table top flat to within and surface finish shall be within	0.000050 inch 16 micro inches TIR
Table periphery runout round table	0.002 TIR
Indexing increments over the entire range	1 degree over the entire 360 degree range

machining operation. The locating pin and sensing dog shall assure indexing accuracy is preserved in the event of an accident.

3.4.8.5 Solid state components. Solid state design shall be used throughout for electronic components.

3.4.8.6 Control circuit voltage. Auxiliary control circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary voltage no greater than 120 volts.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the rotary table shall be not less than those specified in tables I and II for the rotary table size specified.

3.6 Performance. Unless otherwise specified (see 6.2.1), the rotary tables shall be capable of supporting and rotating the weight capacity of "on" center loads. The rotary tables shall be designed for mounting either on primary machine tables or on rigid base suitable prepared for such use. Each motorized feeding table shall be capable of performing all functions specified herein. All rotary tables shall have rotation motion of 360 degrees clockwise and counterclockwise and shall have variable speed, independent of other equipment, such as inspection instruments and workholding devices for use in varied machine operations. Each rotary table shall be capable of performing all the requirements specified herein and tables I and II.

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3.7 Alignment tolerances. Each rotary table shall meet the alignment tolerances specified in and table III.

3.8 Standard equipment. Unless otherwise specified, all equipment normally furnished by the manufacturer on his standard commercial rotary table shall be furnished. When specified (see 6.2.1), required additional equipment shall be furnished and fully described.

3.9 Optional equipment. Optional equipment specified in the following sub-paragraphs shall be furnished with each rotary table as specified, (see 6.2.1).

a. Manual chuck with oversize center hole.
b. Power chuck.
c. Angle plate for vertical operational.

3.10 Repair parts. Repair parts shall be furnished as specified (see 6.2.1).

3.11 Marking on plates and charts. All words on plates and charts shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

3.11.1 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be securely attached to each rotary table. If a chart is furnished, it shall be placed in a transparent plastic folder or permanently sealed between clear plastic sheets with suitable means of mounting. The chart or plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.11.2 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each rotary table. The nameplate shall contain the information listed below. If the rotary table is a special model, the model designation shall include the model of the basic standard rotary table and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Load capacity (lbs)
Table speed (rpm)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

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3.12 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be written in the English language.

3.13 Workmanship. Workmanship of the rotary tables and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination and all tests shall be cause for rejection.

4.5 Quality conformance inspection. A quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), the quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.1, and the

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inspection in 4.8. Failure of the item to pass the examination, the test, and the inspection shall be cause for rejection.

4.6 Examination. The rotary table shall be visually examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Operational test. The rotary table shall be operated at no load for 30 minutes for each type of operation for which it is designed. Proper operation of all controls, motors, adjusting mechanisms, and other accessories shall be verified during the trial period.

4.7.2 Alignment tolerance test. Each rotary table shall be tested to determine that it complies with the alignment tolerance requirements of 3.7.

4.7.3 Performance tests. Performance tests shall be made to determine the positioning accuracy of the rotary table as follows.

4.7.3.1 Positioning table, manual feed. Starting from a reference zero, the accuracy of precision lockbolt positions shall be determined at not less than four positions. Positioning accuracy of precision lockbolt positions shall be within 10 seconds of arc at each lockbolt position.

4.7.3.2 Positioning table, motor feed. The table shall be stopped at not less than eight random positions, and location accuracy of the digital read-out compared with absolute location from reference zero. Location indicated by the scale shall be accurate within 10 seconds of arc at each selected test position.

4.7.4 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.9.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The rotary tables covered by this specification are to be used in shops and depots of activities in the Department of Defense.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

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a. Title, number, and date of this specification.

b. Type, style, and size required (see 1.2).

c. First article, if required (see 3.1).

d. If rotary tables design and construction is required in a specific dimensional system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Additional safety and health requirements, if required (see 3.2.5).

g. Painting, if different (see 3.3.5).

h. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.9).

i. Type of T-slots required for round top tables, parallel or radial (see 3.4.1).

j. Type of taper or centering hole required, if required (see 3.4.1).

k. Flame hardened cast iron ways, if different (see 3.4.2).

l. Type lift assistance required (see 3.4.7.1).

m. Type II worm and worm gear and manual hand crank positioning if different (see 3.4.7.2).

n. Type III motorized positioning system, if different (see 3.4.7.3).

o. Type IV feeding table gear change system, if different (see 3.4.7.4).

p. Type V feeding table reversible, variable speed DC motor drive, if different (see 3.4.7.5).

q. Electrical system and input power, if different (see 3.4.8).

r. Digital readout system, when specified (see 3.4.8.1).

s. Motor enclosure, if different (see 3.4.8.3).

t. Controls, if different (see 3.4.8.4).

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u. Size and capacity, if different (see 3.5).

v. Additional performance features, if required (see 3.6).

w. Standard equipment, if different (see 3.8).

x. Specify and fully describe optional equipment, if required (see 3.9).

y. Specify applicable repair parts, if required (see 3.10).

z. Lubrication chart or plate, if different (see 3.11.1).

aa. Nameplate, if different (see 3.11.2).

bb. First article inspection, if different (see 4.4).

cc. Quality conformance inspection, if different (see 4.5).

dd. If preservation, packing, and marking in accordance with MIL-M-18058 is required, specify level of preservation and packing required (see 5.1).

6.2.2 Contract data requirements. Required technical data such as operator's manuals, parts lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Subject term (key word) listing.

Air lift assistance
Clamping system
Clockwise
Counterclockwise
DC electronic drive
Hand operated positioned system
Hydraulic power system
Hydrostatic assistance
Manual positioned with lift capabilities
Parallel T-slots
Powered feed at variable speeds
Radial T-slots
Self contained units
Silicon control rectifier
Table gear system
Worm and worm drive system

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6.5 Cross-reference of classification changes. The rotary table classification changes from the previous issue of this specification are as follows:

MIL-T-80218	MIL-T-80218A
Style A	Style A
Size 1	Size 1
Size 2	Size 2
Size 3	Size 3
Size 4	(deleted)
	Size 4a (New size)
Size 5	Size 5
Size 6	Size 6
Size 7	Size 7
Style B	Style B
Size 1	Size 1
Size 2	Size 2
Size 3	Size 3
Size 4	(deleted)
	Size 4a (New size)
Size 5	(deleted)
	Size 5a (New size)
Size 6	Size 6
Size 7	Size 7
Size 8	(deleted)
	Size 8a (New size)
Size 9	Size 9
Size 10	Size 10
Size 11	Size 11
Size 12	Size 12
	Size 13 (New size)
	Size 14 (New size)

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AR
Navy - SH
Air Force - 99

Review activities:
Army - AL
Navy - SH
Air Force - 84

User activity:
Navy - MC

Preparing activity:
DLA - IP

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-80218A	Tables, Rotary, Horizontal, Manual and Power Feed

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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